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Comparative analysis of machine learning algorithms for money laundering detection

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Abstract

This study explored the effectiveness of anomaly detection techniques in identifying fraudulent financial transactions, with a particular focus on money laundering activities. The research addressed the growing challenges of financial fraud, which significantly impacts economies and financial institutions by leading to substantial monetary losses and undermining trust in financial systems. This research examined contemporary machine learning (ML) algorithms, including XGBoost, K-Nearest Neighbors, Random Forest, Isolation Forest, and Support Vector Machines, to analyze transaction data for anomalies indicative of fraudulent behavior. This study's approach involves data collecting, system design, implementation, data analysis, and experimental setup. This research aimed to identify robust algorithms for detecting financial fraud. In the results notably, XGBoost shows an output of 1.0, 1.0, 1.0, 1.0, and 0.94 for accuracy, precision, recall, F1 score and AUC respectively to outperform other ML algorithms experimented in money laundering detection. The findings underscore the potential of ML algorithms capability to combat money laundering and indeed enhance anti-money efforts and offer financial institutions a powerful, resource-efficient method for fraud detection in large-scale transaction environments.

Keywords Fraud detection, Money laundering, Machine learning, Anomaly detection, XGBoost

1 Introduction

Financial fraud refers to deceptive practices aimed at securing an unfair or unlawful financial gain [1]. Financial fraud encompasses various deceptive practices, including credit card fraud, insurance fraud, and mortgage fraud, all of which negatively impact economies, causing significant monetary losses and eroding trust in financial system [2, 3]. Financial institutions play a critical role in monitoring these activities as they are often the first line of defence. However, traditional detection methods struggle to keep up with the sophisticated and dynamic nature of financial fraud [4]. There are limited studies focuses on money laundering, which is a process criminals use to make illegally obtained funds appear legitimate, often by breaking up large transactions, creating fake businesses, or using digital currencies. Thus, machine learning (ML) offers a modern



approach, it analyses transaction data and identifying suspicious anomalies, making it a key technology in the fight against financial crime [5–7].

Machine learning (ML) can predict a specific distribution of leading digits in naturally occurring datasets, including anomaly detection tool to flag irregular digit distributions that may indicate fraud. Money laundering is complicated, always changing, and meant to be hard to find, thus the existing techniques fall short in detecting money laundering [8, 9]. This study investigates the application of ML and anomaly detection techniques to improve the detection of money laundering [10]. Machine learning algorithms, such as K-Nearest Neighbours (KNN), Random Forest, Support Vector Machines (SVM), and Isolation Forest, are effective in identifying unusual patterns in transaction data, helping to distinguish between regular and fraudulent activities [11–13]. Specifically, the study aims to comparing ML algorithms for the prospect of a more robust framework that will improve fraud detection accurately and reducing false positives, thus enhancing anti-money laundering efforts in financial institutions such as Banks, Credit Unions, Casinos and more.

The subsequent sections of this study are organized as follows: Sect. 2 contains the literature review of the research addresses fraud detection utilizing ML-based algorithms. Section 3 presents the methodology that delineates the research design and execution environment. The experimental findings and comments present the execution and outcomes in Sect. 4. Finally, Sect. 5 contains the conclusion that synthesizes the study and proposes future research directions.

2 Literature review

This section provides a review of the relevant literatures on financial fraud detection, focusing on the use of ML algorithms on detecting anomalies.

2.1 Fraud detection before the advent of machine learning

Kolhatkar et al., [14] developed a system that receives real-time transaction data from various channels of a payment card system through an API. This allows for immediate analysis of transactions as they occur, which is crucial for timely detection of suspicious activities. They created transactional profiles for each payment card, including the ingress and egress channels and funding sources. This profiling is essential for establishing a baseline of normal behavior against which current transactions can be evaluated. They implemented a predictive algorithm that evaluates incoming transaction data by comparing it to the established transactional profiles. This algorithm calculates a probabilistic money laundering score for each transaction, which helps in assessing the likelihood of money laundering activities. Based on the probabilistic scores and a set of predefined rules, the system generates suspicious activity reports (SARs). These reports recommend whether to approve or report on the transaction, thereby facilitating compliance with regulatory requirements. The system not only transmits suspicious activity reports back to the payment card system but also forwards them to identified regulatory bodies. This ensures relevant authorities are informed of potential money laundering activities promptly.

Moustafa et al., [15] introduced a plan-based framework for anti-money laundering systems (PBAMLS). This framework is innovative and consists of two main phases: a monitoring phase and a planning phase. The monitoring phase employs an automata

approach to detect potential money laundering activities, while the planning phase utilizes STRIPS-based planning to reinforce the findings from the monitoring phase. In the monitoring phase, the authors utilized a deterministic finite automaton derived from regular expressions that define various money laundering processes. This formalism allows for systematic detection of suspicious transactions, minimizing reliance on heuristic methods and human expertise. One of the significant discoveries of the authors is that their framework significantly reduces the number of false-positive alarms, which are common in existing anti-money laundering systems. By employing rigorous formalism, the framework enhances decision-making certainty and decreases the workload on investigators. The authors tested the applicability of PBAMLS through various case studies, demonstrating its effectiveness in real-world scenarios. The framework includes supporting modules for data collection, link analysis, and risk scoring, which further enhances its utility in detecting money laundering cases.

Gasanova et al. [16] analysed data from the Federal Financial Monitoring Service of the Russian Federation, specifically focusing on financial transactions within the banking system. This data included encrypted information about electronic money transfers between legal entities and individuals, which was crucial for the study. The research employed various data mining methods to detect money laundering activities. The author utilized network graph analysis to examine the relationships and interactions between different entities involved in financial transactions. This approach allowed for the identification of suspicious patterns and behaviors indicative of money laundering. One of the key discoveries was the ability to identify suspicious bank accounts and track transactions effectively. The author used centrality measures, such as in-degree and out-degree, to determine the most significant participants in the transaction network. This analysis helped in pinpointing the main actors involved in potential money laundering schemes. They discovered that the analysis of network nodes could reveal clusters of closely related bank accounts, which are often indicative of money laundering activities. By grouping these nodes, the author was able to trace the flow of money and identify patterns that suggest illicit activities. The findings of the research contribute to enhancing the efficiency of financial monitoring systems. By identifying the main actors and their interconnections, the author proposed methods that could greatly enhance the identification and avoidance of money laundering and terrorism financing.

2.2 Machine learning for anti-money laundering frameworks

Monteiro [17] investigated the use of various algorithms for deep learning (DL) and ML to improve the detection of suspicious transactions in cryptocurrencies, specifically targeting the Bitcoin Elliptic dataset. The algorithms included Deep Neural Networks (DNN), RF, KNN, and Naive Bayes (NB). Authors highlighted the challenges faced by traditional AML measures in the digital realm, particularly due to the anonymity and transnational nature of cryptocurrencies, which make them vulnerable to money laundering activities. They designed and evaluated different algorithms to assess their effectiveness in identifying illicit transactions. The study aimed to enhance the detection capabilities of suspicious transactions within the Bitcoin Elliptic dataset. The findings revealed that the RF model outperformed the other algorithms, achieving an impressive overall accuracy of 95%. This model effectively distinguished between legal and illegal transactions while minimizing the risks of overfitting. The study concluded that

the integration of these ML technologies could significantly enhance AML monitoring capabilities in cryptocurrency markets. This would provide reliable decision support for financial institutions and regulatory bodies. The authors suggested that future research could explore more complex DL models or ensemble learning methods to further improve classification accuracy across diverse datasets. They emphasized the need for real-time monitoring of emerging money laundering patterns to strengthen the global AML framework.

Reite et al., [18] utilized distinct bank data specifically from small- and medium-sized enterprises (SMEs). This data was crucial for examining how different models for classifying client risk could predict future suspicious transactions. In order to categorize client risk, the authors investigated a number of client risk data sources and used several ML techniques. Their goal was to determine which models could minimize false positives while accurately predicting suspicious transactions. According to the study, client risk classification models were able to forecast suspicious transactions in the future with high accuracy. Notably, the accuracy of these models was much increased by adding credit score and accounting data.

The improvement enables banks to more effectively weigh the requirement to lower false positives against the danger of overlooking suspicious transactions. According to the authors, banks may become more effective at focusing their AML efforts on the most vulnerable customers by using their strategy, which makes use of easily accessible data sources. Investigating any money laundering operations may become more efficient as a result. In the context of AML, the work is acknowledged as the first empirical investigation of ML in customer risk classification. It demonstrates how incorporating new data sources can greatly lower false positives, improving AML systems overall efficacy.

Gandhi et al., [19] utilized data from the US Treasury's Financial Crimes Enforcement Network (FinCEN) to inform their research. This data served as the foundation for applying various AI/ML methods to enhance AML efforts. They implemented a range of ML and DL algorithms. The authors employed hyperparameter tuning techniques, specifically grid search and randomized search, to optimize the performance of their models. This step was crucial for enhancing the accuracy and reliability of their predictions. The study revealed significant findings regarding the effectiveness of the AI/ML algorithms: The RF classifier achieved an impressive average accuracy of 99.99% in predicted states. The random forest classifier with gradient boosting regressor performed exceptionally well in credit card transaction prediction as well as simultaneous year and state prediction. CNN and MLP demonstrated promise in debit card transaction prediction. With the lowest mean squared error (MSE) of 2.9 and the greatest R-squared (R^2) value of 0.24, the gradient boosting regressor showed a strong capacity for pattern recognition. With respective ROC AUC scores of 0.55 and 0.53, the logistic regression and RF classifiers likewise demonstrated strong performance in credit card transaction prediction.

Strategies for enhancing AML frameworks, include emphasizing interpretability of models, optimizing hyperparameters, developing specialized models, utilizing ensemble methods, implementing data augmentation establishing real-time monitoring systems. They suggested exploring the integration of blockchain technology into AML practices as a potential area for future research and improvement.

Raj et al., [20] focused on how AI can be utilized to enhance AML efforts, specifically addressing the three stages of money laundering: placement, layering, and integration.

They explored the role of AI technologies in each of these stages to improve detection and prevention strategies. The study examined various AI technologies, including ML, Artificial Neural Networks (ANN), and Convolutional Neural Networks (CNN). These technologies were analyzed for their potential to strengthen AML systems, which encompass components. The authors identified several challenges associated with implementing AI in AML processes such as inadequate data availability, the need for a skilled workforce and risks of failure and bias in AI systems. To address the challenges, the authors suggested several strategies, such as implementing a risk-based approach and incorporating KYC data effectively. Ensuring continuous monitoring and human oversight. The study concluded that AI holds substantial potential for improving AML processes. It emphasized that AI can significantly enhance the detection and prevention of money laundering activities, thereby providing financial institutions with more robust tools to combat this global issue.

2.3 Combining anomaly detection with traditional ML algorithms to detect money laundering

Lorenz et al., [21] recognized that traditional supervised learning methods are ineffective due to the scarcity of labelled data in the context of money laundering detection. They aimed to address this issue by exploring alternative methods that could work with minimal labels. Initially, the authors tested existing unsupervised anomaly detection methods. These methods are designed to identify outliers in datasets without any labelled instances. However, they found that these methods were inadequate for detecting illicit patterns in real Bitcoin transaction datasets. The anomalies detected did not reliably indicate money laundering activities, highlighting the limitations of relying solely on unsupervised techniques. Overcoming the limitations of unsupervised methods, the authors proposed an active learning solution. Active learning is an iterative process where the model queries a human expert to label specific instances, thereby gradually increasing the amount of labelled data available for training. This approach allows the model to learn from a small subset of labelled data while still leveraging the vast amount of unlabelled data available. The authors conducted experiments comparing their active learning approach against a fully supervised baseline. Remarkably, they found that their active learning method could achieve performance comparable to the fully supervised model using only 5% of the labels. This result demonstrates the effectiveness of active learning in scenarios with limited labelled data, making it a practical solution for real-world applications in anti-money laundering (AML) efforts. The study utilized a real-life Bitcoin transaction dataset released by Elliptic, emphasizing the importance of using authentic data for reliable conclusions. The findings indicated that experiments on synthetic data could be misleading, reinforcing the need for real-world datasets in AML research.

Mishra et al., [22] carried out a deep comparison of various ML algorithms, including KNN, RF, NB, DNN, and evolution metrics. They aimed to identify which of these techniques are most effective in detecting suspicious transactions. Findings on Effectiveness: Through their analysis, the authors discovered that utilizing a combination of these ML algorithms can significantly strengthen the ability of banks and other financial institutions and government agencies to find and stop money laundering. This multi-faceted approach is essential for protecting the integrity of the financial system. The work

contributes to the broader field of financial security by providing insights into ways to use ML algorithms to improve.

detection capabilities. The comparative analysis serves as a valuable resource for practitioners looking to implement or improve their anti-money laundering strategies.

Bellei et al., [23] introduced a novel dataset, Elliptic2, which contains 122,000 labelled subgraphs of Bitcoin clusters. This dataset is crucial for training ML algorithms to recognize patterns associated with illicit activities, as it provides a rich source of data for learning the “shapes” that money laundering exhibits in cryptocurrency transactions. The authors posited that money laundering activities create distinct subgraphs that can be identified through their unique features. They aimed to develop ML algorithms capable of detecting these anomalous patterns, which are characterized by the flow of funds from illicit to licit clusters without a change of ownership. The authors employed Graph Neural Networks (GNNs) to analyse the subgraphs. GNNs are particularly suited for this task as they can capture relational information at a subgroup level, allowing for a more nuanced understanding of the connections between nodes (Bitcoin addresses) involved in transactions. This approach enables the model to learn from the structure of the data rather than just individual transactions. The primary objective was to classify subgraphs as either suspicious (indicative of money laundering) or not. This binary classification task leverages the labelled data in the Elliptic2 dataset to train the models, allowing them to generalize and make predictions on previously unseen subgraphs. The authors experimented with various subgraph classification methods, including GNN-Seg, Sub2Vec, and GLASS. These methods utilize different strategies for embedding and analysing the subgraphs, with GLASS being noted for its ability to incorporate background graph information, which may provide additional context for detecting anomalies.

Flayyih et al., [24] aimed to clarify what Benford’s Law is and how it functions in the context of financial analysis. This law is a statistical principle that predicts the frequency distribution of the first digits in numerical data, which can be used to identify anomalies in financial records. The research highlighted how Benford’s Law serves as a crucial tool for auditors and forensic accountants in detecting financial manipulation. The author emphasized its importance in identifying fraudulent activities, particularly in income reporting and other financial items within companies. The author explored the digital analysis aspect of Benford’s Law, detailing how it can be applied to various financial datasets. This analysis helps in spotting irregularities that may indicate fraudulent behaviour. The literature review included an examination of significant studies that have previously addressed the application of Benford’s Law in fraud detection. The author synthesized findings from these studies to underscore the effectiveness of this law in various financial contexts, such as joint-stock companies and sales manipulation. The author concluded that Benford’s Law is more than a valuable statistical method for auditors but also a reliable tool for detecting fraud across multiple financial operations. The findings suggest that there should be a greater focus on utilizing this law in auditing practices to enhance fraud detection efforts.

Antonio [25] aimed to determine how deviations from the expected distribution could indicate potential financial irregularities. The authors collected a comprehensive sales dataset consisting of 3,098 records from the Bandung branch and 539 records from the Surabaya branch for the year 2022. The Bandung and Surabaya branches represent distinct operational units. Certain data exhibiting abnormalities were excluded from the

analysis. The authors applied Benford's statistical test to the sales data. They found discrepancies between the expected Benford probability distribution and the observed probabilities, suggesting the presence of errors and fraudulent activities. The research uncovered multiple instances of fraud in the Surabaya branch, where salesmen manipulated prices to benefit personally. Specifically, four fraud cases were identified, with sales prices set far below the appropriate amounts. Customers confirmed that they had given money to the salesmen to cover these price discrepancies. The study also revealed errors in price inputs, such as transposition errors where the prices entered the system differed from those on the sales orders. The above work as well as emphasized the need to combine it with other data analysis and auditing techniques to ensure a thorough investigation of financial data [26].

3 Methodology

This section focuses on the detailed research and analysis methods that were undertaken in this study to gather all the research information. The research methodology is explained in this section, it includes the data collection, system design, implementation, data analysis and experimental setup. This study implements a qualitative research design focusing on the application of ML algorithms for anomaly detection in financial transactions for money laundering [2]. The research is experimental, aiming to evaluate the effectiveness of various anomaly detection techniques in identifying fraud. The rationale behind this design as shown in Fig. 1 is to systematically test and compare different ML algorithms to determine their accuracy and efficiency in fraud detection.

3.1 Data collection

The data for this study consists of real-world financial transactions datasets. The data is taken from IBM, it is also available on Kaggle (<https://www.kaggle.com/code/ebaahossam/anti-money-laundering-detection-using-pyspark>). This data contains analysis that is perfect for detecting financial crimes such as money laundering. Since it covers transaction amounts, time, accounts and types of currency, this data is suitable to be used for anomaly detection, clustering and classification models to identify unusual patterns that could indicate laundering or other fraudulent activities. The data preprocessing procedures were implemented to make sure that all the missing data is handled, standardize currencies and encode categorical variables. Since the study aims to find the perfect ML algorithms to detect money laundering in financial transactions, we incorporated contemporary ML algorithms that are best known for detecting fraud. The pre-processed data is divided into training and testing sets. This ensures that each subset maintains the same proportion preventing any skewedness or biased representation. The dataset was a balanced dataset which means a categorical dataset where all classes (or categories) in the target variable are equally represented or have approximately the same number of examples. Therefore, there was not need for Undersampling, Oversampling, Class weights and Overfitting of the models.

3.2 System design

The fraud detection system collects and pre-processes transaction data, handling missing values, extracting relevant features, and standardizing currencies. It employs an anomaly detection engine using ML algorithms like XGBoost, KNN, RF, SVM and



Fig. 1 Methodology Framework

Isolation Forest to identify anomalies and flag suspicious transactions in real time. The system splits data for training and testing, evaluating models with metrics like accuracy and precision, and fine-tunes parameters for optimal detection. A user interface provides a dashboard for visualizations and alerts analysts to high-risk transactions. Deployed on a scalable cloud platform, the system regularly retrains models to adapt to evolving fraud patterns, effectively combining traditional and ML algorithms for robust fraud detection.

3.3 Implementation environment and experimental setup

This section outlines the practical execution of the research methodology. The experiments were conducted in a cloud-based computing platform google collab using python programming language and its libraries. The hardware specifications for ML Training in anomaly detection of this study using a windows 11 computer running Intel® core™ @ 3.00 GHz and 32 RAM, GPU NVIDIA A100 / V100 / RTX 3090, Networking 10–100

Gbps Ethernet. Google Collab has a runtime allocation of 12 GB RAM and 107 GB storage. Independent variables are the features extracted from transaction data, and the dependent variable is the binary classification of transactions as fraudulent or non-fraudulent. The models are trained using 80% of the data, and the remaining 20% is reserved for testing. To ensure consistency, all models undergo the same preprocessing and training procedures. Performance is measured using statistical metrics, and visual tools are employed to assess the detection capabilities of each model.

3.4 Data analysis

The study employs descriptive statistics and visualization techniques to explore transaction patterns. Models are evaluated based on metrics such as accuracy, precision, recall, and F1 score, with particular attention to the false positive rate, which is a common challenge in fraud detection models. ROC curves and confusion matrices are used to visualize model performance and compare different algorithms. These metrics serve unique purposes and give insight into different aspects of the model's effectiveness. Figure 2 is a line graph showing the number of transactions that were deposited from the bank and also to the bank from 2022-09-01 to 2022-09-17. The line graph shows an uneven distribution of the transactions, the transactions are at their peak from the 2022-09-01 and drastically decrease. They increase from the 2022-09-08 and drastically decrease again and stay consistent onwards. This trend is very alarming, and it may indicate data manipulation.

In Fig. 3 we have a correlation heatmap of the amount received and amount paid. With the correlation of 0.74 between amount paid and amount received it indicates a strong functional relationship between the two variables. Also, high correlation might be a results of data quality issues.

Figure 4 is a boxplot of all the currencies used in the dataset and the amount received. The Boxplot shows that most anomaly transactions were sent using Rupee, Yen and Mexican Peso respectively. Since all these currencies have visible outliers.

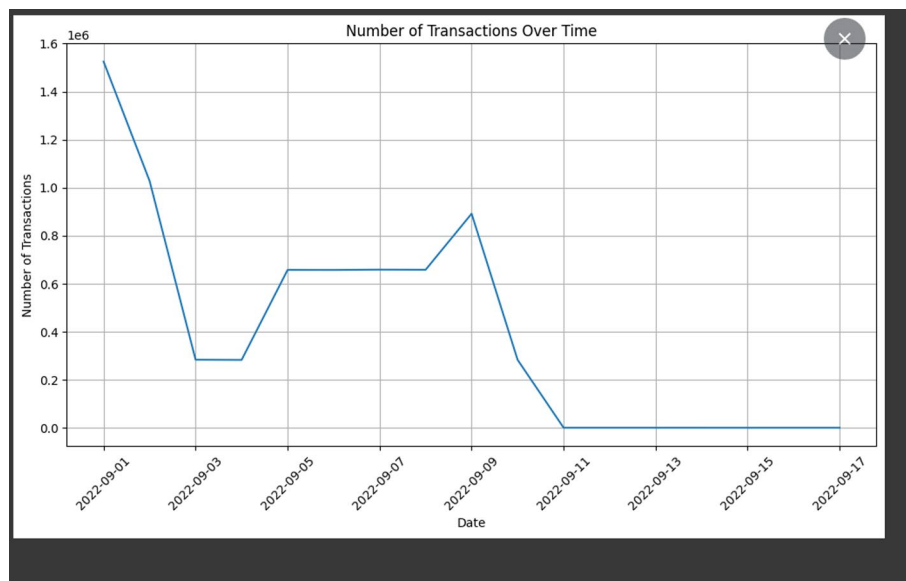


Fig. 2 Line graph for transactions per year

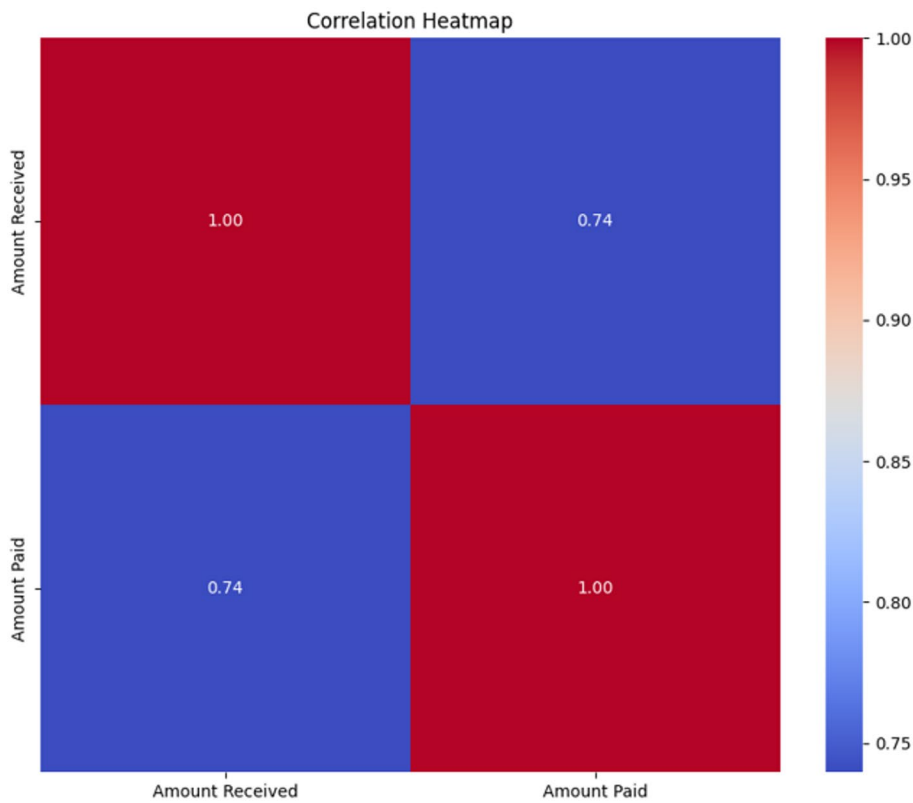


Fig. 3 Heatmap of the amount received and amount paid

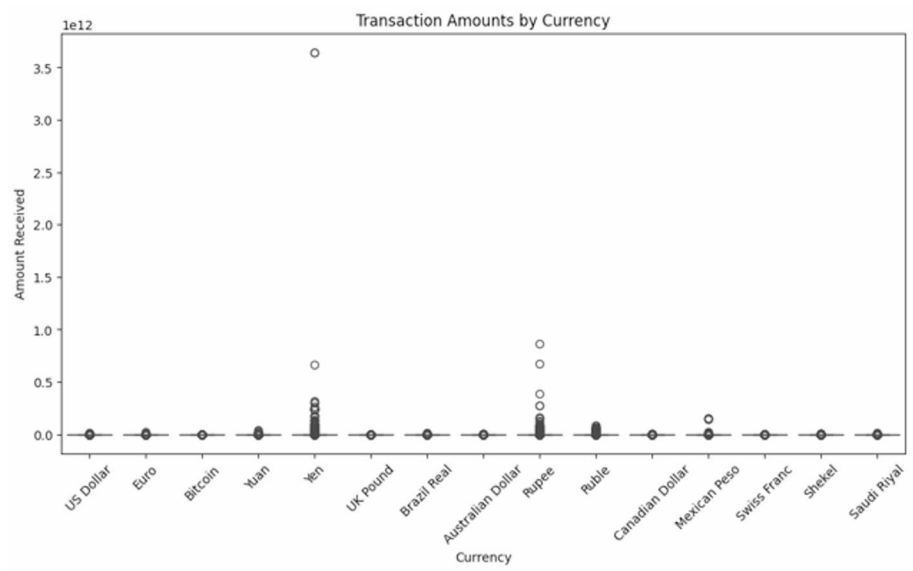


Fig. 4 Boxplot of the currency received

The distribution of payments is depicted in the pie chart that can be found in Fig. 5, and it appears that the majority of transactions were actually completed. ACH, credit card, and check payments.

4 Results and discussion

This section presents the experimental results of the proposed solution. The classifiers used were XGBoost, KNN, RF, Isolation Forest and SVM. This shows a comparison of the ML algorithms to the data distribution of the study turned out to be like. The terms for the algorithm evaluation were based on the state-of-the-arts measure such as accuracy, F1 scores, recall, precision, confusion matrix and ROC.

4.1 Results and implications

A True Negative is when the model accurately predicts the negative class. A False Positive happens when the model wrongly says that an instance is positive when it is actually negative. A false negative happens when the model wrongly predicts that an instance is negative when it is actually positive. The comprehensive and comparative analysis of the results shown in Table 1 and the graphical illustration that summarizes a model's prediction results on a classification problem in Fig. 6. It shows on the account of the state-of-the-arts metrics in this domain.

Accuracy.

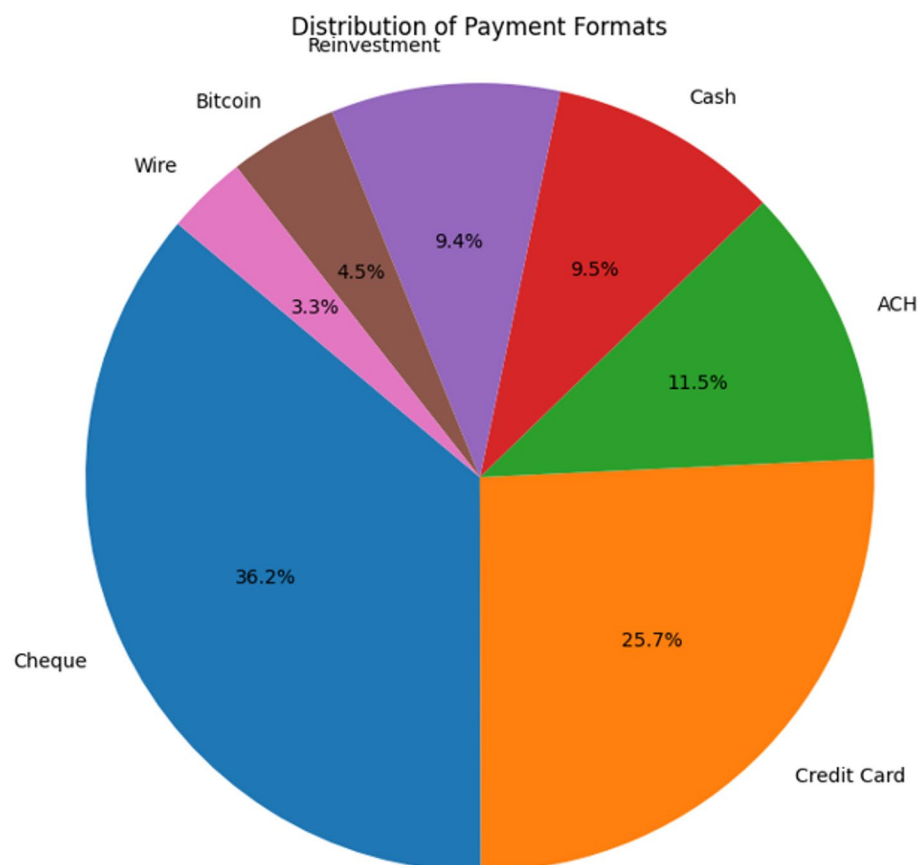
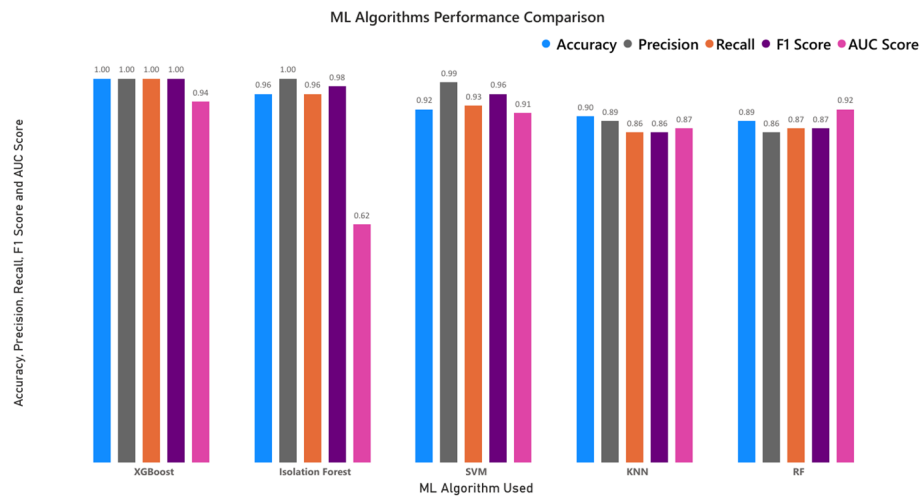


Fig. 5 Pie Chart for payments distribution

Table 1 ML model results

ML algorithm used	Accuracy	Precision	Recall	F1 score	AUC
XGBoost	1.00	1.00	1.00	1.00	0.94
KNN	0.9014	0.89	0.86	0.86	0.87
RF	0.89	0.86	0.87	0.87	0.92
Isolation forest	0.96	1.00	0.96	0.98	0.62
SVM	0.92	0.99	0.93	0.96	0.91

**Fig. 6** Comparative analysis of ML Algorithms for Money Laundering Detection

The accuracy of a model indicates how well it identifies both fraudulent and legitimate transactions. In this study, XGBoost has the highest accuracy score of 1.00, indicating that it accurately classified all transactions in the dataset. This perfect accuracy suggests that XGBoost demonstrated an exceptional ability to distinguish between fraudulent and legitimate transactions. Following XGBoost, Isolation Forest reached an accuracy of 0.96, indicating strong performance with only a small margin for error. SVM also performed well, with an accuracy of 0.92, suggesting it was generally effective at correctly classifying transactions. K-Nearest Neighbours (KNN) and RF had accuracy scores of 0.9014 and 0.89, respectively, which, while still strong, indicate slightly lower levels of accuracy. Overall, the accuracy results reveal that XGBoost was the best-performing model in accurately identifying transactions, setting a high benchmark for the other models. True Positive is an instance where the model correctly predicts the positive class. Accuracy is the ratio of correctly predicted observations to the total observations, and the formula for accuracy experimental results metrics was based on Eq. 1.

$$Accuracy = \frac{True\ positives + True\ Negatives}{Total\ Observations} \quad (1)$$

Precision.

Precision is a critical metric for fraud detection, as it indicates how well the model avoids false positives which is a common challenge in identifying fraudulent activity. The XGBoost and Isolation Forest achieved perfect precision scores of 1.00, meaning that every transaction they identified as fraudulent was indeed fraudulent. These results show that these two models excelled in minimizing false positives, which is particularly

valuable for financial institutions that aim to avoid overwhelming investigators with false alarms. SVM performed very well with a precision of 0.99, indicating it made very few false positive errors and remained highly reliable in detecting true fraudulent transactions. KNN and RF showed lower precision scores, at 0.89 and 0.86, respectively, suggesting a higher rate of false positives. The high precision scores of XGBoost, Isolation Forest, and SVM demonstrate their strength in producing accurate alerts, making them effective tools in fraud detection. Precision is the ratio of correctly predicted positive observations to the total predicted positive observations and the formula for Precision experimental results metrics was based on Eq. 2.

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \quad (2)$$

Recall.

Recall measures the model's ability to capture actual fraudulent cases, indicating how effectively it avoids missing true fraud instances. XGBoost has the perfect recall score of 1.00, capturing all actual fraudulent transactions without missing any. This perfect recall score underscores XGBoost's strength in identifying all cases of fraud, ensuring no fraudulent transaction went undetected. Isolation Forest has a recall score of 0.96, indicating it was also highly effective at detecting true fraudulent transactions. SVM performed comparably well with a recall of 0.93, further supporting its reliability. KNN and Random Forest, with recall scores of 0.86 and 0.87 respectively, showed solid performance but slightly lower sensitivity in identifying all fraudulent transactions. Thus, high recall scores in XGBoost, Isolation Forest, and SVM highlight their effectiveness in detecting true positives, a crucial aspect in preventing fraud from slipping through undetected. Recall is the ration of correctly predicted positive observations to all observations in the actual class and the formula for Recall experimental results metrics was based on Eq. 3.

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \quad (3)$$

F1 Score.

The F1 score provides a balanced metric by combining both precision and recall, thus offering insight into how well each model manages the trade-off between identifying true positives and avoiding false positives. XGBoost again achieved a perfect F1 score of 1.00, indicating a flawless balance between high recall and precision. This result underscores XGBoost's robustness as a well-rounded model, excelling in both capturing fraudulent transactions and minimizing false alerts. Isolation Forest also demonstrated a high F1 score of 0.98, showing that it maintained both high recall and precision. SVM followed with a strong F1 score of 0.96, reflecting its capability to balance accuracy and reliability. In contrast, KNN and Random Forest yielded lower F1 scores of 0.86 and 0.87, respectively, suggesting they had a less optimal balance between capturing true fraud cases and minimizing false positives. Overall, the F1 scores emphasize that XGBoost, Isolation Forest, and SVM were the most balanced models, supporting their use in high-stakes fraud detection. F1 score is the harmonic mean of precision and recall, providing

a balance between the two and the formula for F1 score experimental results metrics was based on Eq. 4.

$$F1Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (4)$$

Area under the curve (AUC).

AUC measures the model's ability to distinguish between fraudulent and legitimate transactions across different classification thresholds. XGBoost and Random Forest produced the highest AUC scores of 0.94 and 0.92, respectively, showing strong discrimination between classes and confirming their reliability across varied thresholds. SVM achieved a robust AUC of 0.91, further supporting its effectiveness in distinguishing transaction types. KNN followed with an AUC of 0.87, indicating a moderate capacity to separate fraudulent and legitimate transactions. Isolation Forest, however, recorded a notably lower AUC of 0.62, suggesting that, while effective in specific metrics, it may have struggled to consistently distinguish between classes across a range of thresholds. The confusion matrix and ROC curve results shown in Figs. 7 and 8, highlight XGBoost's consistent performance across various thresholds, underscoring its robustness in distinguishing between transaction classes. Table 1 depicts the performance results of ML algorithms. In the results, XGBoost show an output of 1.0, 1.0, 1.0, 1.0, and 0.62 for accuracy, precision, recall, F1 score and AUC respectively to outperform other ML algorithms experimented in money laundering detection. Table 2 is comparison of existing studies and this ensures a validation of the effectiveness of each ML algorithm in the current study.

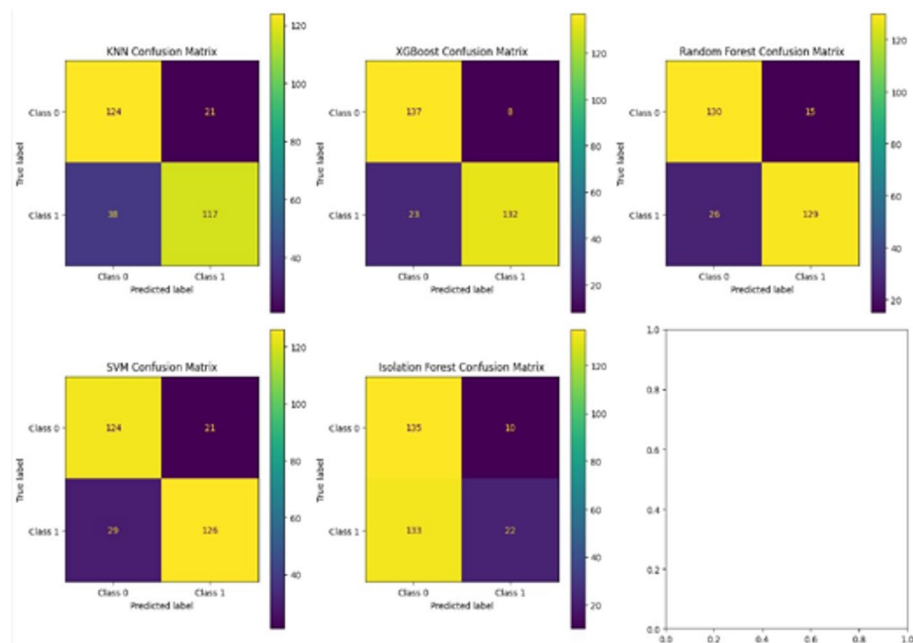


Fig. 7 Confusion Matrix

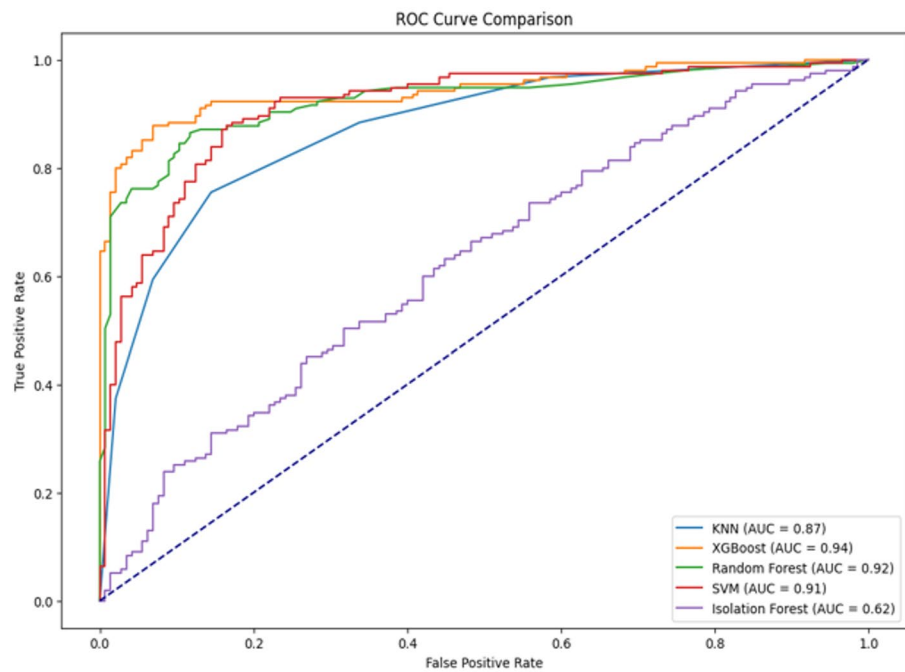


Fig. 8 ROC Curve

Table 2 Comparison of existing studies and current study

Author	Method	Findings
Monterio [17]	The algorithms included Deep Neural Networks (DNN), RF, KNN, and Naive Bayes (NB).	The findings revealed that the RF model outperformed the other algorithms, achieving an impressive overall accuracy of 95%. This model effectively distinguished between legal and illegal transactions while minimizing the risks of overfitting.
Gandhi [19]	They implemented a range of ML and deep learning (DL) algorithms.	The study revealed significant findings regarding the effectiveness of the AI/ML algorithms: The random forest classifier achieved an impressive average accuracy of 99.99% in predicted states. With respective ROC AUC scores of 0.55 and 0.53, the logistic regression and random forest classifiers likewise demonstrated strong performance in credit card transaction prediction.
Antonio [25]	The authors applied Benford's statistical test to the sales data and found discrepancies between the expected Benford probability distribution and the observed probabilities	The research uncovered multiple instances of fraud in the Surabaya branch, where salesmen manipulated prices to benefit personally. Specifically, four fraud cases were identified, with sales prices set far below the appropriate amounts.
Antunes [27]	Their goal was to evaluate how effectively this law can be utilized in identifying fraudulent activities.	The research concluded that Benford's Law is a powerful analytical tool for identifying anomalous data. It was cost-effective and allows organizations to focus their efforts on further analysis of instances that deviate from expected patterns. The authors suggested that Benford's Law should be used as an initial screening tool in the fraud detection process. By applying it early on, it can help identify areas that require more detailed investigation using more robust analytical methods.
Current study	The ML algorithms used are XGBoost, KNN RF Isolation Forest and SVM	This study compares ML approaches, providing a novel, multi-dimensional framework. XGBoost outperformed metrics, and anomalies, enabling more reliable and scalable fraud detection.

4.2 Contribution of the research

Current study achieved higher accuracy with XGBoost and demonstrated better precision, recall, and F1 score, indicating a more reliable classification system and this has advance beyond [17]. Current study surpassed in providing more balanced metrics (AUC > 0.9 for XGBoost and SVM), showing robust performance across all evaluation metrics, this is an improvement [19]. Some study used Benford's Law on financial transactions and money laundering detection which are inefficient and inadequate like ML algorithms which offers a comprehensive and hybrid approach with broader applicability beyond money laundering [2, 3]. The current study contributes to the body of knowledge beyond [25, 27] as it demonstrates the capability and strengths of ML algorithms for enhanced fraud detection with the other studies observed and compared to the current study. It is observed that in this study RF isn't the strongest classifier as it did in the other studies, but the model that excelled is XGBoost, with the accuracy of 1. The result opined the following studies [28–30].

5 Conclusion

The study concluded that leveraging advanced anomaly detection methods can substantially strengthen anti-money laundering efforts and improve overall fraud detection capabilities within financial institutions. This research aimed to enhance fraud detection in financial transactions with ML algorithms. Five ML algorithms XGBoost, KNN, RF, Isolation Forest, and SVM were evaluated using metrics such as accuracy, precision, recall, F1 score, and AUC. The hybrid approach demonstrated high effectiveness, with XGBoost and SVM outperforming RF in accuracy and reliability, the study offered a cost-effective way to flag broad anomalies. The results highlight the potential of ML algorithms for robust fraud detection. Future research could explore integrating traditional and statistical anti-money laundering techniques such as Benford Law with ML algorithms to further enhance anomaly detection accuracy in financial transactions. Authors will also consider cloud-based deployment platform for practical insights in the future work.

Abbreviations

AML	Anti-Money Laundering
API	Application Programming Interface
KNN	K-Nearest Neighbors
ML	Machine Learning
PBAMLS	Plan-Based Anti-Money Laundering System
ROC	Receiver Operating Characteristic
SAR	Suspicious Activity Report
STRIPS	Stanford Research Institute Problem Solver
SVM	Support Vector Machine
XGBoost	Extreme Gradient Boosting

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Author contributions

Conceptualization, and methodology, S.A.A., SM, and P.M. writing—original draft, software development and data curation, visualization and validation SM and SAA; study administration, resources, editing and review, supervision SAA and P.M. All authors have read and agreed to the published version of the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

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